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A pediatric ECG database with disease diagnosis covering 11643 children

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Electrocardiogram (ECG) is a common non-invasive diagnostic tool for cardiovascular diseases. Adequate data is crucial in utilizing deep learning to achieve intelligent diagnosis of ECG. The existing ECG datasets almost only focus on adults and most of them do not provide cardiovascular disease diagnosis. In this study, we propose an ECG database with cardiovascular disease diagnosis for children aged 0-14 years old. This dataset is acquired from 11643 hospitalized children at the First Affiliated Hospital of Zhengzhou University from 2018 to 2024, including 14190 pediatric ECG records, of which 12334 were 12 lead and 1856 were 9 lead. The sampling rate is 500 Hz and the record length is 5-120 seconds. We followed the recommendations of AHA/ACC/HRS and the diagnostic statements in the consensus of Chinese ECG experts to encode and convert all ECG records. In this dataset, 3516 ECG records were diagnosed with cardiovascular diseases, and these labels were derived from 19 common diseases in the pediatric cardiovascular field, including myocarditis, cardiomyopathy, congenital heart disease, and Kawasaki disease.

Background & Summary

According to the 2024 World Heart Report released by the World Heart Federation (WHF), cardiovascular disease mortality ranks first globally. According to statistics, over 10 million people in the United States have died from cardiovascular disease in the past decade[1]. Meanwhile, many studies suggest that the onset of cardiovascular disease begins in childhood and adolescence, and can even be traced back to infancy[2–5].

Electrocardiogram (ECG) is a convenient tool for detecting and recording cardiac electrical activity in clinical medicine, as well as an important method for diagnosing cardiovascular diseases and early screening[6–8]. In recent years, deep learning has achieved significant success in medical diagnosis and has been widely applied to the automatic classification of cardiac abnormalities in ECG signals[9]. There have been many related studies. However, most of these studies used a universal ECG model and did not differentiate the study subjects into adults and children.

There are significant differences between children and adults in clinical medicine in the field of cardiovascular disease[10]. Firstly, there are different manifestations of the ECG. Children undergo significant anatomical and physiological changes during various stages of growth and development, underscoring the significance of age differences, children's ECG is substantially divergent from adult ECG. The larger the age gap, the greater the divergence. Specifically, this is manifested in heart rate, intervals, lead waveforms, and amplitude time[11,12]. Secondly, the incidence of cardiovascular diseases varies. Common pediatric cardiovascular diseases include myocarditis, cardiomyopathy, congenital heart disease, arrhythmia, Kawasaki disease, etc. Adult cardiovascular diseases mainly include hypertension, coronary heart disease, valve disease, etc. Therefore, targeted research on pediatric ECG, combined with pediatric cardiovascular diseases, is of significant preventive, diagnostic, and therapeutic importance.

At present, the existing ECG datasets in the world, such as the famous MIT-BIH Arrhythmia dataset[13,14], AHA Arrhythmia dataset[15], CSE dataset[16], PTB Diagnostic ECG dataset[17,18], PTB-XL dataset[19,20], as well as the newly proposed Shaoxing People's Hospital dataset[21,22], SPH dataset[23,24], etc., are mainly based on adult ECG and most of them only contain ECG diagnostic statements and do not include disease diagnosis. The PHN

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Name	Patients	Age	Records, leads	Records (Children aged 0-14)	Length	Sampling rate	Disease diagnosis
MIT-BIH arrhythmia dataset[13,14]	47	23-89	48, 2 leads	0	30 min	360 Hz	N
AHA arrhythmia dataset?	N/A	N/A	154, 2 leads	N/A	180 min	250 Hz	N
CSE dataset[16]	1220	N/A	1220, 15 leads	N/A	30 second	500 Hz	N
PTB diagnostic dataset[17,18]	294	17-87	549, 15 leads	0	2 min	1000 Hz	Y
PTB-XL dataset[19,20]	18885	0-95	21837, 12 leads	41	10 second	400 Hz	N
Shaoxing People's Hospital dataset[21,22]	10646	4-98	10646, 12 leads	197	10 second	500 Hz	N
SPH dataset[23,24]	24666	10-100	25770, 12 leads	0	10-60 second	500 Hz	N
PHN dataset[25]	2400	0-18	2400, 12 leads	1778	10 second	$>$ 500 Hz	N
FDA pediatric ECG dataset[26]	40000	N/A	N/A, N/A	40000	N/A	N/A	N
MIMIC-IV-ECG dataset[27]	161352	12-101	800035, 12 leads	5	10 second	500 Hz	Y
ZZU pECG dataset	11643	0-14	12334, 12 leads 1856, 9 leads	14190	5-120 second	500 Hz	Y

Table 1. ECG datasets comparison.

dataset[25] and FDA pediatric ECG dataset[26] are mainly designed for pediatric electrocardiograms, however, the number of PHN datasets is relatively small, and the FDA dataset lacks publicly accessible data sources. The MIMIC-IV-ECG dataset[27] contains a massive amount of ECG data with disease diagnostic labels, but only 5 cases were found in children aged 0-14. The specific comparative information is shown in Table 1.

In this study, we proposed a pediatric ECG database with disease diagnosis labels, the data is sourced from the First Affiliated Hospital of Zhengzhou University from January 2018 to May 2024 (hereinafter referred to as "ZZU pECG dataset"). An important contribution of the ZZU pECG dataset is the 14190 pediatric ECG records from 11643 children aged 0-14 (6740 males, 4903 females). This is currently the largest known open-source pediatric ECG dataset. The sampling frequency of ECG records is 500 Hz, and the recording time is 5-120 seconds. Include basic information about the patient, including age, gender, and ECG diagnosis. Unlike previous datasets, we included patients' disease diagnoses and referred to the 10th edition of the International Classification of Diseases (ICD-10) developed by the World Health Organization[28] for coding conversion to further investigate the potential relationship between pediatric ECG and cardiovascular disease.

We referred to the work of the SPH dataset[23,24] and converted the collected ECG diagnostic statements into standardized diagnostic statements recommended by AHA/ACC/HRS (i.e., the American Heart Association, the American College of Cardiology, and the Heart Rhythm Society) in 2009 (hereinafter referred to as "AHA standard")[29]. Although the AHA standard is the most authoritative worldwide, its data sources are mainly from Western populations, while the ZZU dataset is mainly from Eastern populations. Considering the development of the ECG field and population differences, We further refer to the standardized diagnostic statements recommended by experts in the field of ECG in China to convert the ECG diagnostic statements (hereinafter referred to as "CHN standard")[30].

Methods

Data acquisition. This study has been reviewed and approved by the Ethics Committee for Scientific Research and Clinical Trials at the First Affiliated Hospital of Zhengzhou University (2024-KY-0221-003 and 2025-KY-0369-001). It has been agreed to waive the signing of informed consent forms and to desensitize the ECG data for public scientific research.

The raw ECG data of the ZZU pECG dataset come from hospitalized children at the First Affiliated Hospital of Zhengzhou University from January 2018 to May 2024. The ECG acquisition device and model is MedEx MECG-300, with a 24-bit resolution A/D converter in mV, acquisition time of 5-120 seconds, and sampling frequency of 500 Hz. When collecting ECG data, the physician first issues an order for an ECG examination for hospitalized children, then connects the leads to the child's body, completes data collection after a period of time, and uploads it to the MedEX ECG system. Then, a junior physician in the electrocardiographic room will make a conclusive diagnosis, which will be reviewed by a senior physician. Consensus is considered the final diagnosis; If there is a disagreement, the reviewing physician shall report and discuss it before making the final decision.

In order to pay more attention to pediatric ECG and pediatric cardiovascular diseases, we consulted doctors with senior professional titles in cardiovascular medicine, cardiovascular surgery, pediatric medicine, pediatric surgery, pediatric intensive care, and electrocardiographic room, and set the following rules:

1. Age. In this study, we defined the age of pediatric subjects as 14 years of age or below. Besides, we provided the original age of all children in days. In the conversion and statistical scenario of this manuscript, one year is defined as 365 days, and one month is defined as 30 days. However, the most accurate unit is still in days.
3. ECG leads. Notably, it was usually intractable toput all 6 chest leads on such a young child's chest due to the insufficient cooperation and incomplete development[31]. Based on the advice of pediatric cardiologists, we believe that Children aged 7 years and above should have a complete 12 lead ECG, which includes I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6, Children under the age of 7 who lack V2, V4, and V6 leads on their ECG are also allowed to be represented by V1, V3, and V5 leads. There are 1856 ECG records in the ZZU pECG dataset with 9 leads.
4. Disease diagnosis labels. There are hundreds of diseases in the original medical records. We particularly

Disease Type	Disease Category	ICD-10 Code	Count
Myocarditis	Fulminant myocarditis	(F) I40.0	83
	Viral myocarditis	(V) I40.0	296
	Acute myocarditis	I40.9	70
	Myocarditis	I51.4	186
Cardiomyopathy	Dilated cardiomyopathy	I42.0	86
	Hypertrophic cardiomyopathy	I42.2	24
	Cardiomyopathy	I42.9	18
	Noncompaction of the ventricular myocardium	Q24.8	22
Kawasaki disease	Kawasaki disease	M30.3	194
Congenital heart disease	Ventricular septal defect	Q21.0	1062
	Atrial septal defect	Q21.1	936
	Atrial septal defect (Foramen ovale)	(FO) Q21.1	593
	Atrial septal defect (Ostium secundum defect)	(OSD) Q21.1	95
	Atrioventricular septal defect	Q21.2	61
	Tetralogy of Fallot	Q21.3	89
	Stenosis of right ventricular outflow tract	Q22.1	24
	Patent ductus arteriosus	Q25.0	348
	Pulmonary stenosis	Q25.6	18
	Pulmonary valve stenosis	I37.0	81
Other diseases	(See attribute dictionary file and ICD-10 version[28])	(See attribute dictionary file)	10474

Table 2. Overview of disease type and category.

focus on cardiovascular disease labels in Table 2, while keeping other disease labels without separate counting. The ICD codes for other diseases can be obtained from the attribute dictionary file and ICD-10 version[28].

The data acquisition process is as follows.

1. Retrieve all hospitalized cases of children aged 0-14 from January 2018 to June 2024 using the Neusoft hospital information system. The inpatient case contains the main medical information of the patient during their hospitalization period. The patient ID, date of birth, gender, admission date, discharge date, and disease diagnosis (Excluding outpatient diagnosis) are extracted from the case and exported as an Excel file.
2. Based on the patient ID in the Excel file, retrieve all ECG records from the MedEx ECG system and export them as an XML file. This file contains the patient ID, ECG values, ECG acquisition date, and ECG diagnostic statements.
3. Design a tool to extract information from XML files other than ECG values and convert it into CSV files.
4. Match patient ID in Excel and CSV files, clean ECG records with the same patient ID that do not match the admission and discharge dates, and ensure that all remaining ECG records are from the patient's hospitalization period.
5. According to the disease diagnosis in the Excel file, all cases of myocarditis, cardiomyopathy, Kawasaki disease, and congenital heart disease were screened and retained. The specific disease information is shown in Table 2.
6. Extract the date of birth from the Excel file and the ECG acquisition date from the XML file and calculate the patient's age provided in days. The advantage is that the date calculation in the Python library has built-in leap year logic, so the calculated children age is the most accurate.
7. Through the above steps, we collected all pediatric ECG records that meet the requirements, as well as the patient ID, gender, age, acquisition date, ECG diagnostic statement, and disease diagnose. Finally we selected 3516 ECG records with cardiovascular disease labels and 10674 ECG records with other disease labels.

It is the fact that The MedEx ECG system contains patient's age, However, the age here is based on the admission date, not the collection date. For very younger patients, this error increases. Therefore, it is most reasonable and accurate for us to calculate the patient's age by using the date of birth on the medical record homepage and the acquisition date of the ECG record.

Data processing. In the real data collection, different noises such as Gaussian noise and baseline wander noise, power line interference, and muscle artefacts corrupt the ECG wave all through its receipt and transmission[32]. The MedEx MECG-300 machine used in this study has a built-in filtering function, which allows experienced physicians or nurses to collect ECG signals from patients in a resting state. The quality of the remaining ECG signals is generally good and meets the requirements for use, without the need for additional filtering and denoising. Only ECG records that exclude lead detachment and lack key information such as patient ID, age, diagnostic statements, and ECG acquisition date need to be excluded. The diagnostic statements of the original ECG are in Chinese and manually entered by doctors from the electrocardiographic room. Due to the subjective differences among different doctors, conclusions of the same category also show significant differences in diagnostic statements and require unified coding conversion. The specific steps are as follows:

1. Organize all diagnostic statements and codes in AHA and CHN standards.
2. With the help of a physician, try to include all brief and colloquial statements of these diagnostic statements as much as possible.
3. According to the statement design tool obtained in the previous step, use wildcard characters to extract key characters and convert all diagnostic statements of ECG into diagnostic statements and codes corresponding to AHA and CHN standards. For diagnostic statements that do not have corresponding codes found in AHA and CHN standards, we will retain them in text form.
4. Two cardiologists checked the diagnostic statements of all ECG, and retained ECG records that were converted correctly and met AHA or CHN standards; Correct ECG records with incorrect conversion of diagnostic statements to comply with AHA or CHN standards; Discuss and perform secondary diagnosis on ECG records that cannot be corrected or do not meet both AHA and CHN standards after correction, and make a comprehensive judgment on whether to retain the record.

The AHA standard has six parts, namely primary diagnostic terms, secondary diagnostic terms, modifying vocabulary, concise comparative terms, specifications for the individual and combined application of the above standards, and commonly used combination terms. The core part is the primary diagnostic terminology. Differed form the AHA standard, the CHN standard only has the primary diagnostic statement, so its comprehensiveness and scope of application are lower than the AHA standard, and it only serves as a supplement to the ECG diagnostic statements for the Eastern population. During the encoding conversion process, three types of codes from the AHA standard were used, namely primary diagnostic statements (Nondescriptive statements, convey clinical meaning without additional statements), secondary diagnostic statements (Provides additional statements that can be used to expand the specificity and clinical relevance of both descriptive and other primary diagnostic statements. These secondary statements are divided into 2 groups : "suggests" and "consider") and modifiers (It does not change the meaning of the core statement, but is used to refine the meaning). The conversion of CHN standard only uses the primary diagnostic statement.

In order to protect the privacy of patients, we desensitized the data and regenerate the patient ID, ensuring consistency with the patient's ECG records, disease diagnosis, and other information. Simultaneously set a random integer offset (the specific numerical range is not disclosed) in the acquisition date. If a patient is hospitalized multiple times for ECG acquisition or multiple ECG records are collected during one hospitalization, the order of ECG examination remains unchanged.

The disease diagnosis labels of hospitalized children are obtained from the medical record homepag, which information has been reviewed and does not require secondary verification. We converted all disease diagnoses to ICD-10 codes under the guidance of experienced clinical doctors and disease coding personnel based on the ICD-10 version : 2019 released by WHO[28]. Due to the granularity of ICD-10 code not being detailed enough to give all diseases unique labels, some cardiovascular disease codes in Table 2 are duplicated, and we have set unique identifiers for them to distinguish them. For example, the ICD codes for Fulminant Myocarditis and Viral Myocarditis are (F) I40.0 and (V) I40.0, respectively.

After the above steps, 14190 ECG records containing patient information and disease diagnosis were finally obtained. Using ECG records as the statistical unit, the number of ECG records with cardiovascular disease and non cardiovascular disease were 3716 and 10474, respectively, as shown in Table 2. These cardiovascular disease labels cannot be calculated through single summation, as a portion of the ECG records have one or more cardiovascular disease labels simultaneously.

Data Records

The dataset includes pediatric ECG records, attribute dictionary file, disease code file, ECG diagnostic statement code file and code for reading files, which can be obtained through figshare[33]. All children's ECG signals are saved in Waveform Database (WFDB) format, a standardized file format derived from the WFDB python open-source library, which enables efficient and flexible processing and analysis of biomedical signals. The naming of the data file is associated with the patient ID. If a patient has multiple ECG records, the acquisition order should be attached (such as P00001_E01, P00002_E02, P000002). The attribute dictionary file, disease code file, and ECG diagnostic statement code file will be stored in CSV format. It should be noted that the "," are used to separate fields and ";" are used to separate multiple values within a field "in the CSV file.

The attribute dictionary file includes file name, ECG ID, patient ID, age, gender, acquisition data, sampling point, lead, AHA code, CHN code, ICD-10 code, pSQI, basSQI, bSQI, as shown in Table 3. The age here is based on ECG acquisition date, so there may be multiple ECG records of different ages for the same patient. The acquisition data is based on Beijing time and belongs to the UTC/GMT + 08:00 time zone.

The disease code file displays the disease labels and related ICD-10 codes contained in the ECG records. The ECG diagnostic statement code file displays all ECG diagnostic statements in the ZZU pECG dataset with corresponding codes according to AHA and CHN standards, as well as relevant quantity statistics, see Supplementary Table 1. The ECG records in the ZZU pECG dataset contain a total of 78 primary statements and 13 secondary statements. The number of normal and abnormal ECG records were 2641 and 11549, respectively.

Figure 1 shows the age distribution of pediatric ECG in the dataset, with age groups referenced from the work of Andras Bratinchsak Chieko *et al*[34]. The male-to-female ratio recorded in the ECG is 1.34:1. Figure 2

Attributes	Type	Description
File name	String	Relative path of ECG records
ECG ID	String	Unique ID for ECG record
Patient ID	String	Unique ID for patient
Age	String	Age on the day of ECG acquisition (provided in days)
Gender	String	Gender (MALE,FEMALE)
Acquisition date	String	The date of patient ECG acquisition
Sampling point	Integer	The number of sampling point
Lead	Integer	Number of leads in ECG (12, 9)
AHA code	String	Encoding of diagnostic statements recorded in ECG (converted according to AHA standards)
CHN code	String	Encoding of diagnostic statements recorded in ECG (converted according to CHN standards)
ICD-10 code	String	International Code for Disease Diagnosis (10th Revision)
PSQI	String	PSQI score for each lead(see Technical validation Section)
BasSQI	String	BasSQI score for each lead(see Technical validation Section)
BSQI	String	BSQI score for each lead(see Technical validation Section)

Table 3. Attributes in attribute dictionary files.

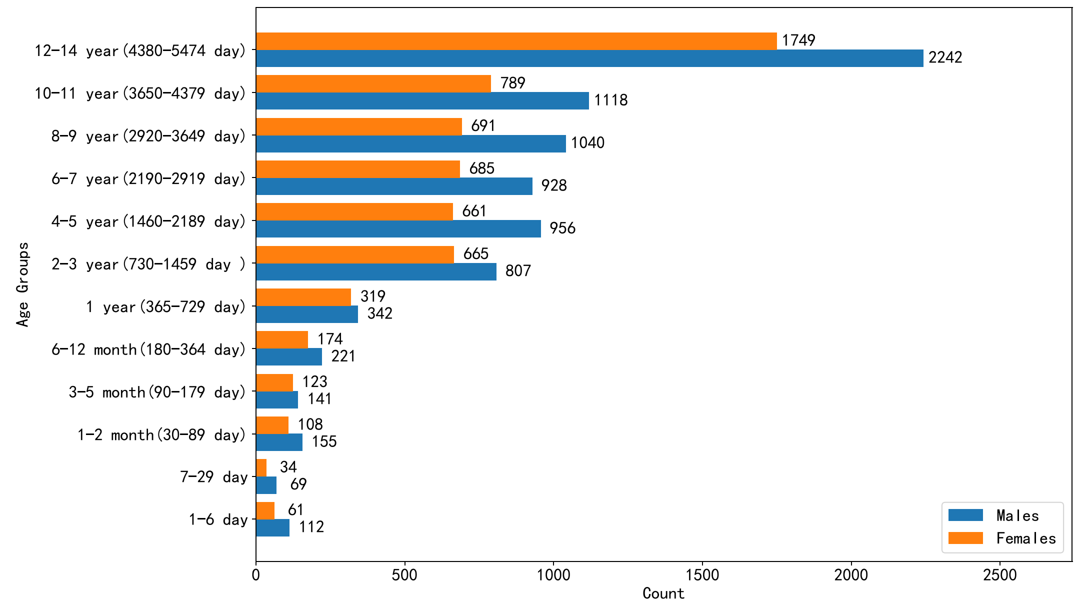

Fig. 1 Overview of patient age and gender.

shows the quantity and gender distribution of these ECG records under the labels of myocarditis, cardiomyopathy, Kawasaki disease, and congenital heart disease. Figure 3 further illustrate the age distribution of these ECG records.

Table 4 shows the length of ECG signal records in the ZZU pECG dataset, with 92.91% of ECG records concentrated in the 20-40 seconds interval, which is in line with the recommendation of the electrocardiographic room for static ECG acquisition time greater than 20 seconds to capture more comprehensive abnormal ECG signals. Tables 5 and 6 show the number of ECG records with cardiovascular disease labels and all disease labels in the dataset. There are 13.97% of ECG records have more than one cardiovascular label, and 64.9% of ECG records have more than one disease label. Table 7 shows the number of ECG diagnostic statements included in the ECG records in the dataset. It can be seen that 49.80% of the ECG records in the dataset contain more than one ECG diagnostic statement. Table 8 shows the number of ECG records included in the dataset, with 14.52% of patients having more than one ECG record.

Technical Validation

To verify the quality of ECG records, we used three signal quality indices (SQI), pSQI (QRS Power Spectrum Distribution SQI), basSQI (Baseline Relative Power SQI) and bSQI(beat detection signal quality index), to evaluate the quality of raw ECG signals[35–37].

PSQI is used to evaluate the degree of influence of high-frequency noise in ECG signals, its standard and identification criteria is shown in equation (1) and (2) ($P(f)$ represents the power spectral density (PSD) of the ECG signal, l_1, l_2 and l_3 vary slightly with the heart rate shown in equation (3)). Due to the fact that the heartbeat

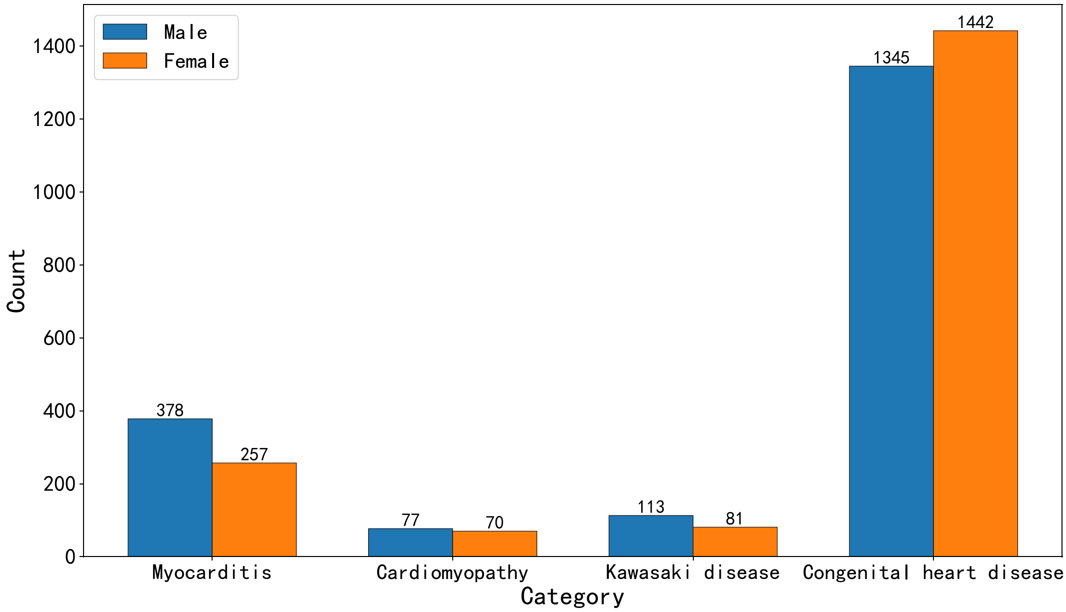

Fig. 2 Distribution of gender in ECG records with cardiovascular disease labels.

cycle is mainly composed of important characteristic vectors such as P wave, QRS complex, T wave, etc., the QRS wave accumulates 99% of the energy of the ECG signal and is the most stable. The energy of QRS complex is concentrated in the frequency band centered around 10 Hz, with a width of 10 Hz. Low pSQI means the presence of high-frequency noise interference in the [15,40 Hz] frequency band, while high pSQI means that noise containing the same frequency in the [5,15 Hz] frequency band, in addition to the ECG signal itself, increases the power of the signal.

$$pSQI = \frac{\int_{f=5Hz}^{f=15Hz} P(f)df}{\int_{f=5Hz}^{f=40Hz} P(f)df} \tag{1}$$

$$\text{ECG}\begin{cases}\text{optimal}, & \text{pSQI} \in [l_1,\ l_2];\\ \text{suspicious}, & \text{pSQI} \in [l_3,\ l_1];\\ \text{unqualified}, & \text{pSQI} > l_2, \text{ or pSQI} < l_3.\end{cases} \tag{2}$$

$$\begin{cases}\in [60\text{bpm},\ 130\text{bpm}], & l_1 = 0.5,\ \ l_2 = 0.8,\ \ l_3 = 0.4;\\ \in [130\text{bpm},\ 160\text{bpm}], & l_1 = 0.4,\ \ l_2 = 0.7,\ \ l_3 = 0.3;\end{cases} \tag{3}$$

BasSQI is used to evaluate the degree of low-frequency noise interference caused by respiration in ECG signals, its standard and identification criteria is shown in equation (4) and (5). Low basSQI means that the power in the [0, 1 Hz] frequency band is abnormally high relative to the power in the [0, 40 Hz] interval, and the data has a baseline abnormal offset and poor quality.

$$basSQI = \frac{1 - \int_{f=0Hz}^{f=1Hz} P(f)df}{\int_{f=0Hz}^{f=40Hz} P(f)df} \tag{4}$$

$$\text{ECG}\begin{cases}\text{optimal}, & \text{basSQI} \in [0.95,\ \ 1],\\ \text{suspicious}, & \text{basSQI} \in [0.9,\ \ 0.95],\\ \text{unqualified}, & \text{basSQI} < 0.9.\end{cases} \tag{5}$$

BSQI is defined as the ratio of beats detected synchronously by both algorithms in the ECG to the total number of beats detected by either algorithm, as shown in equation (6). The identification criteria of bSQI are given as equation (7).

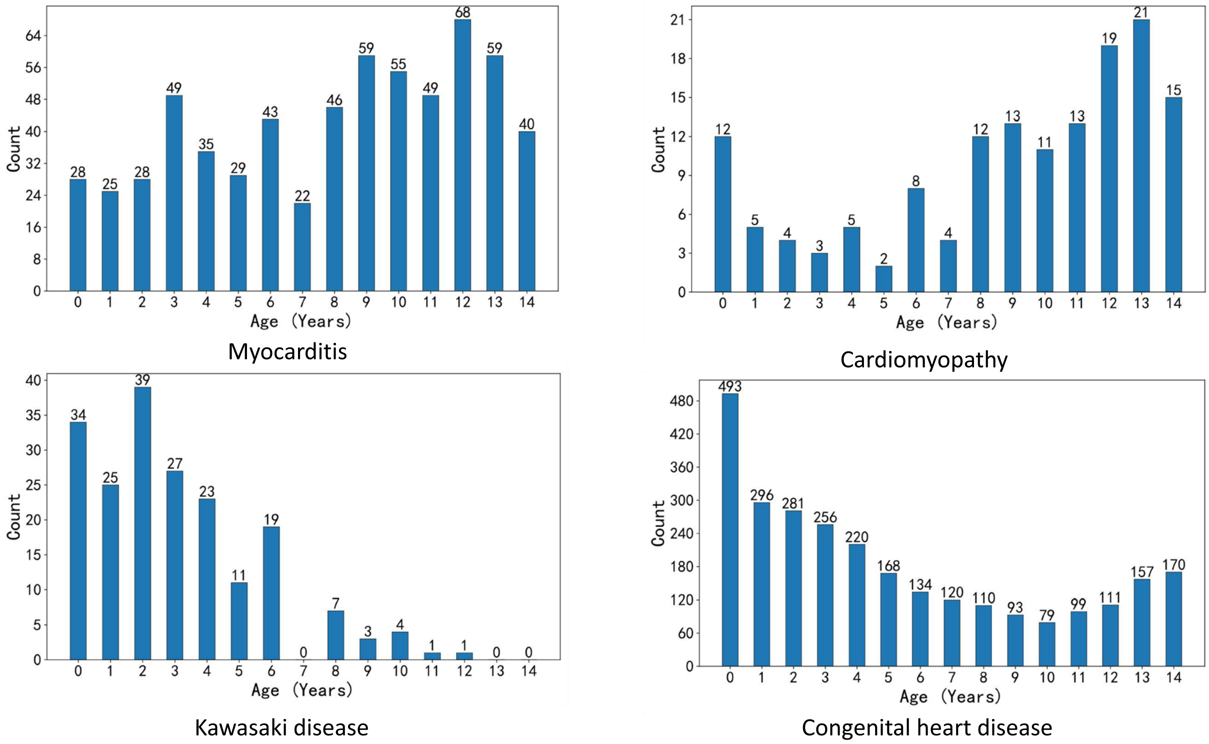

Fig. 3 Distribution of age in ECG records with cardiovascular disease labels.

Seconds	[5, 10)	[10, 20)	[20, 30)	[30, 40)	[40, 60)	[60, 80)	[80, 100)	[100, 120)
ECG records	7	927	3736	9448	31	34	3	4

Table 4. Overview of ECG records time.

Number of disease labels (19 types of cardiovascular diseases)	1	2	3	4
ECG records	3197	470	47	2

Table 5. Overview of Disease Diagnosis and ECG record.

Number of disease labels (all)	1	2	3	4	5	6	7	8	9	10	11
ECG records	4983	3422	2046	1218	705	441	333	243	220	164	415

Table 6. Overview of Disease Diagnosis and ECG record.

ECG diagnostic statements	1	2	3	4	5	6	7	8	9	10	11
ECG records	7123	4080	1602	755	365	119	75	39	18	4	10

Table 7. Overview of diagnostic statement per ECG record.

ECG records	1	2	3	4	5	6	7	8	9	10	10+
Patients	9952	1327	202	66	33	18	9	7	11	9	9

Table 8. Overview of ECG records per patient.

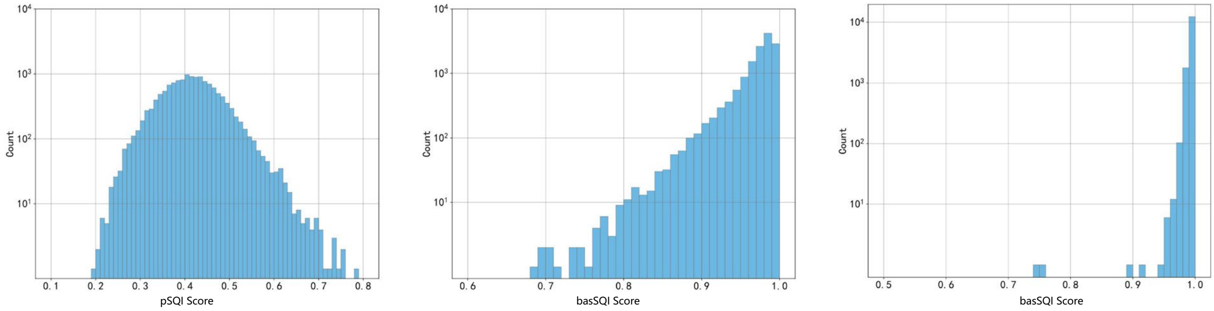

Fig. 4 Distribution of pSQI, basSQI and bSQI score in our dataset.

$$bSQI(k) = \frac{N_{\text{matched}}(k,\ w)}{N_{\text{all}}(k,\ w)} \tag{6}$$

Where $N_{\text{matched}}(k, w)$ is the number of beats agreed upon by both the Nabian algorithm and the Promac algorithm[38] (within 150 ms) and $N_{\text{all}}(k, w)$ is the number of all beats detected by either algorithm (without double counting the matched beats).

$$ECG\begin{cases} optimal, & bSQI > 90\%; \\ suspicious, & bSQI \in [60\%,\ 90\%]; \\ unqualified, & bSQI < 60\% \end{cases} \tag{7}$$

For each piece of data, we first calculated the pSQI, basSQI and bSQI indices for each lead separately and then took the average. The distribution of results is shown in Fig. 4. Finally, we put the pSQI, basSQI, and bSQI scores of the ECG leads into the attribute dictionary file.

Usage Notes

Each ECG record in this dataset is stored as a WFDB type, consisting of two files: header file [.hea] and data file [.dat]. The header file [.hea] records the data format, lead information, sampling frequency, patient gender, age, and disease type of the signal through ASCII code storage. The data file [.dat] stores signals in binary format, with a value stored every two bytes, and each value is 16 bits in size, store in little-endian format.

The ZZU pECG dataset proposed in this study mainly comes from the pediatric population and includes a portion of children with cardiovascular disease. Compared to previous datasets, we have added cardiovascular disease diagnostic labels (sourced from the patient's electronic medical record homepage) to the ECG records of children with cardiovascular disease. These labels are the most common pediatric cardiovascular diseases in the real world (excluding arrhythmia). It should be noted that static ECG belongs to transient electrophysiological signals, and even patients with cardiovascular disease may have a period of time when the ECG is normal; At the same time, the patient's hospitalization period is not always in a state of illness, but a dynamic process from incubation period to onset period, treatment period, and then to recovery period. Without accurately determining the specific treatment stage and disease state of the patient, we still ensure that each ECG record is generated during hospitalization. The final ECG dataset may contain abnormal disease diagnoses accompanied by normal ECG diagnosis results. To ensure the authenticity of the data, we have retained these data.

This dataset can be used for various classification tasks, such as detecting and classifying abnormal ECG records and cardiovascular diseases in children. However, due to the uneven distribution of labels, we strongly recommend only classifying labels with a large amount of data. In the follow-up work, we will also enrich and supplement the dataset.

Code availability

The code for reading the attribute dictionary file and ECG signal data, ECG dataset segmentation and ECG signal quality indices can be obtained from figshare[33].

Received: 7 October 2024; Accepted: 19 May 2025;
Published: 26 May 2025

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Acknowledgements

This project has received funding from the Henan Provincial Health Commission and Henan Provincial Department of Science and Technology, with project ID LHGJ2024025 and 241100310200. Thank you for the support from the electrocardiographic room, Pediatric Department, Cardiovascular Department, and Cardiology Department of the First Affiliated Hospital of Zhengzhou University. Thank you to Beijing MedEx Technology Co., Ltd. for their support in medical devices and data acquisition, and thank you to Neusoft Group and Beijing Jiahe Meikang Information Technology Co., Ltd. for their support in diagnostic information software.

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Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41597-025-05225-z.

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